Offline: data and training

Expert labels
DENTEX + Tufts, FDI notation

Teacher VLM + real tools
label-directed LangGraph loop
tiered perturbation of the shown box

none	small	big
45%	25%	30%

Cross-family verifier
label check, repair, decontamination

Verified corpus (3,694 traces)
1,847 with tools + 1,847 without

1 · SFT
verified traces

2 · GRPO
GT-anchored reward

harness off

rules + skills

LoRA adapters
SFT, GRPO off, GRPO rules + skills

frozen SFT adapter = KL reference

Extension 1: rules and skills harness

rules.md
versioned system rules

Skill library (5 skills)
index in prompt; read_skill loads a body on demand

Inference loop

prompt

read_skill

panoramic X-ray

Policy
Qwen3.5-9B
+ LoRA adapters
thought, then tool call

call

obs

adapters

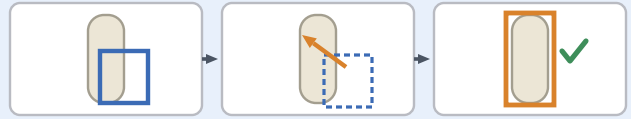
findings JSON

decode

Tools (run on the native image)

zoom_crop	window_level
denoise	enhance_contrast
contralateral_compare	locate_tooth
fdi_label	**nudge_crop**

grounding is a hypothesis



Extension 2: speculative decoding

MTP draft
fine-tuned on target

A B C D

Target
(policy) verifies

keep A, B; target corrects C

Evaluation

- Test: DENTEX validation, 50 images, once
- Dev: ~70 training images, for selection
- Pooled set-matched F1, bootstrap CIs
- Probes: tool removal, box displacement

data / model · our contribution · verified output · optional extension · trained · frozen